

THE UNIVERSITY OF WESTERN AUSTRALIA

Computer Science and Software Engineering

SEMESTER 1, 2015 EXAMINATIONS

CITS4403
Computational Modelling

FAMILY NAME:_____ GIVEN NAMES: _____

STUDENT ID: | | | | | | | | | SIGNATURE:_____

This Paper Contains: **15** pages **(including title page)**
Time allowed: **2:10 hours (including reading time)**

INSTRUCTIONS:

There are 23 questions. Marks for this paper total 120, Candidate must answer all questions.

Answers are to be written on this question paper in the spaces provided.

Calculators are not permitted.

PLEASE NOTE

Examination candidates may only bring authorised materials into the examination room. If a supervisor finds, during the examination, that you have unauthorised material, in whatever form, in the vicinity of your desk or on your person, whether in the examination room or the toilets or en route to/from the toilets, the matter will be reported to the head of school and disciplinary action will normally be taken against you. This action may result in your being deprived of any credit for this examination or even, in some cases, for the whole unit. This will apply regardless of whether the material has been used at the time it is found.

Therefore, any candidate who has brought any unauthorised material whatsoever into the examination room should declare it to the supervisor immediately. Candidates who are uncertain whether any material is authorised should ask the supervisor for clarification.

Supervisors Only - Student left at:

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(Questions 1 to 22 carry 5 marks each)

1. Explain how to construct a regular graph of size n and degree k when k is even (and less than n)?

[5 marks]

2. Explain what a random graph is and how to generate one.

[5 marks]

3. Give an algorithm for determining if a graph is connected or not.

[5 marks]

4. Give the order of growth of bisection search, and explain your answer?

[5 marks]

5. Explain how the Hashtable add operation can take constant time on average despite the need to resize the map.

[5 marks]

6. Explain how to implement a Breadth First Search through a graph.

[5 marks]

7. Describe two different explanations for the Small World Phenomenon.

[5 marks]

8. Explain how the "cliquishness" of a graph can be measured by a clustering coefficient.

[5 marks]

9. How would you characterise the typical main differences between a computational modelling explanation for a phenomenon compared to a traditional scientific explanation?

[5 marks]

10. What is Zipf's law?

[5 marks]

11. Explain how to use "preferential attachment" to create random graphs with a distribution of degrees that obeys a power law.

[5 marks]

12. Describe at least one major objection to Popper's claim that falsifiability demarcates science from pseudo-science.

[5 marks]

13. Define one-dimensional Cellular Automata.

[5 marks]

14. Explain the four classes of behaviour shown by Cellular Automata.

[5 marks]

15. In the Game of Life can there be long-lived patterns? Explain your answer.

[5 marks]

16. Explain how a Cellular Automaton can produce an object of dimension about 1.56.

[5 marks]

17. Explain with an example the difference between Realism and Instrumentalism.

[5 marks]

18. Define Self-organized criticality.

[5 marks]

19. What does it mean to say that a 10% fall in the stock market may have no explanation?

[5 marks]

20. Explain how Schelling's modelling is supposed to show that segregation of neighbourhoods arises without racism.

[5 marks]

21. Explain what the iterated prisoner's dilemma is.

[5 marks]

22. Discuss the claim that Emergence is just ignorance.

[5 marks]

23. Briefly review a chapter of a popular non-fiction book that you have read recently covering a topic related to complexity science.

[10 marks]

END OF PAPER

Blank page for extra writing if needed